

Formal Record of Dissertation Meeting

Student's Responsibility

It is the student's responsibility to take notes of the meeting and forward them to the supervisor in a timely manner.

Student:	Mykhailo Shpyl	Supervisor:	Dr. Vini Vijayan
-----------------	----------------	--------------------	------------------

Programme:	MSc Artificial Intelligence and Big Data Analytics
Date:	9 June 2026 (Meeting 2)

Key Points Discussed (must include timetable for research and training record):
Presented the architecture and progress. Walked through the full pipeline using the diagram and slide deck: detection (hybrid DeBERTa plus stylometric features), explainability, argument mining, question generation (two backends), Bloom's classification, and the output guide. Supervisor read the materials beforehand and was positive on the work. Stylometric features explained. Discussed using text predictability (perplexity, via next-word probability) and burstiness (variation in sentence length and structure across a document) as signals, since AI text tends to be more predictable and more uniform than human writing. Explainability and ablation. Explained that SHAP, Integrated Gradients, and attention show which passages drove a flag, and that faithfulness is checked by ablation: removing the highlighted evidence and confirming the detector's confidence actually changes. Agreed the system works on assumptions and the explanations make those assumptions inspectable. Dataset work. Presented the cleaned and balanced BAWE sample (640 essays): stratified by disciplinary group and native/non-native status, per-student cap to avoid over-representing one writing style, and a student-level train/validation/test split to prevent style leakage. Non-native writers oversampled because they tend to be flagged as AI more often. Supervisor guidance: avoid over-structuring the data. Vini advised not to spend too much time manually balancing the dataset. An AI system should be able to learn from less curated data, and heavy manual structuring risks an unrealistic result. Move on to training rather than further balancing. Supervisor guidance: structured vs unstructured comparison needs two models. If demonstrating the difference between training on the structured sample versus less structured data, this requires two separate models run in parallel, not one. A model already trained on the structured set cannot be fairly tested on the unstructured set. Three-class as a possible extension. Confirmed starting with two classes (human vs AI). If that works well, a third "partial AI" class could be added later as an extension. Partial AI is expected to overlap with both other classes, so it stays out of the first phase. Evaluation approach confirmed. Question-generation evaluation: a discrimination simulation (context-aware vs context-blind model answering each question, the score gap showing how well a question targets the specific passage), plus LLM-as-judge with Krippendorff's alpha for agreement across models, and standard precision/recall for detection. No human participants. Assessment format (viva). Viva now looks likely to be included alongside the presentation; to be confirmed with the coordinators. Vini recommends preparing as if there will be a viva, and considers it a positive opportunity to discuss the work. Conference abstract suggested. Vini encouraged submitting an abstract (200 to 250 words) to an NLP or AI conference in the next few months (Ireland or elsewhere, hybrid is fine). Conference route preferred over

a journal given the 13-week timeline. An accepted abstract would strengthen the final submission.

Indicative timetable: Foundation (data, environment) essentially complete. Next: generate the length-matched AI essays and train the detector once HPC access is confirmed; build explainability and argument mining; then question generation and Bloom's; then integration and evaluation; writing throughout. Introduction chapter to be written at outline level now, with tools and methods filled in after the model is built.

Action items for Student:

- Stop further manual balancing of the dataset; proceed to generating the AI essays and training the detector.
- Run the detector on the balanced sample locally to get a first result while HPC access is confirmed.
- If pursuing the structured vs unstructured comparison, design it as two separate parallel models.
- Write the Introduction chapter at outline level only for now (no detailed tools/methods until the model is built).
- Search for NLP/AI conferences (Ireland or Europe, hybrid acceptable) with abstract deadlines in the next few months, and prepare a 200 to 250 word abstract.
- Fix the percentage rounding on the disciplinary-group chart before using it in the dissertation (the four group percentages did not sum to 100).
- Upload this meeting record to the OneDrive folder and email Vini once uploaded.
- Complete the application form referenced at the end of the meeting.

Action items for Supervisor:

- Confirm access to the ATU HPC compute (check whether the test machine is currently available) and revert with a date.
- Confirm the assessment format (viva alongside presentation) with the coordinators.
- Review the uploaded materials before the next meeting if time allows.

Date of next meeting:

Tuesday 16 June 2026, 12:00 (online)

Issues requiring immediate attention:

HPC compute access. Still the main timeline dependency. AI-essay generation at scale and detector training need GPU; supervisor is confirming availability. Local run on the sample proceeds in the meantime.

Chart fix. The disciplinary-group percentages on one figure do not sum to 100 and must be corrected before inclusion in any documentation.